

Problem Sheet 1

1.

(a) Explain what the terms “saturation” and “offset” mean when applied to a sensor or instrument. By what physical mechanisms might the response of the following systems suffer from either or both effects?

- (i) A mercury thermometer;
- (ii) A mechanical pressure gauge;
- (iii) A Hall effect probe.

(b) What methods can you suggest to eliminate any zero-offset effects through calibration?

Note: you may need to look up the physical principles underlying these sensors.

2.

A mercury thermometer consists of a spherical bulb with a fixed volume V and a capillary tube with a bore of radius r into which the mercury can expand. The coefficient of linear expansion of a substance is typically defined by:

$$\alpha = \frac{1}{L} \frac{dL}{dT},$$

where L is the linear dimension. For mercury, $\alpha = 1.82 \times 10^{-4} \text{ K}^{-1}$.

(a) Show that the rate of change of mercury volume with temperature is $3V\alpha$. (Note that the volume of mercury inside the bore is negligible compared to that inside the bulb, hence its expansion can be neglected.)

(b) If l is the length of the mercury column in the bore, show that the static sensitivity of the thermometer is

$$\frac{dl}{dT} = \frac{3\alpha V}{\pi r^2}.$$

(c) What bore radius is needed in order to achieve $1\text{mm}/^\circ\text{C}$ static sensitivity with a 1cm^3 bulb?

(d) The total length of the bore is 60 mm. Suggest a suitable maximum/minimum measurement range for general purpose use. If the mercury is replaced with coloured ethanol (6 times larger expansion coefficient compared to mercury) discuss the impact on the thermometer's performance parameters (range, saturation, etc). How might we establish the calibration of the thermometer and adjust for any zero-offset?

3.

Note: For this question you'll need the AD590 data sheet. You would not be expected to decipher a manufacturer's data sheet as part of an exam – but you are encouraged to look a few up in any case.

An AD590 temperature sensor built into an “H” type package is to be used to measure the temperature of a large vat of stirred oil. If the sensor is initially at room temperature, and is plunged rapidly into the oil at a temperature of about 100°C , how long does it take to reach 86.5% of the steady state reading? After 14 seconds, estimate the overall error (assume the calibration error had not been trimmed out beforehand).

4. SEMINAR QUESTION

A PMT with 8 dynodes is set up with each dynode connected to the next by a series of $100\text{ k}\Omega$ resistors. A high-voltage supply operating at 1.1 kV is used, with the negative supply terminal connected to the photocathode and the positive terminal grounded. The first dynode (nearest to the photocathode) is connected to the HV supply through a $1\text{ M}\Omega$ resistor. The other end of the dynode chain is grounded via the final $100\text{ k}\Omega$ resistor. The PMT has an anode to collect the electrons, which is connected to ground via a $50\text{ }\Omega$ resistor. The photocurrent is determined by measuring the voltage across this resistor. The photocathode has a quantum efficiency of 35% and is illuminated by green light from a continuous-wave laser operating at a wavelength of $0.54\text{ }\mu\text{m}$. Each photon *that is detected* produces a single photoelectron from the photocathode, and each electron that strikes a dynode plate releases a total of 7 additional electrons.

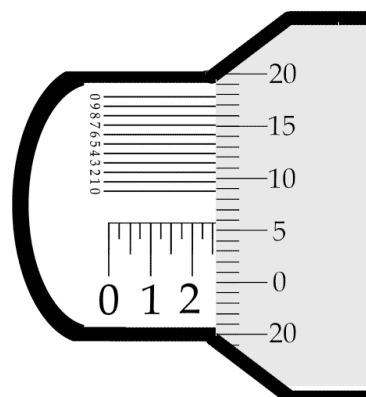
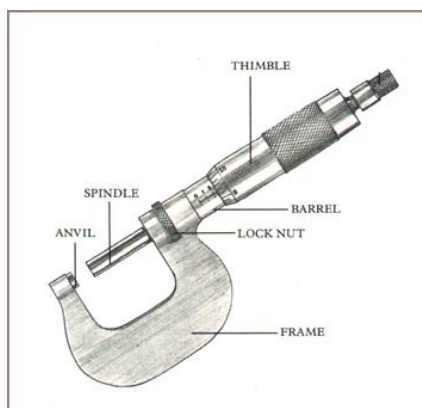
- Draw a schematic of this experimental setup;
- What is the average charge liberated by the final dynode for each photon incident on the photocathode?
- For what continuous-wave optical power does the photocurrent saturate the last stage of the dynode chain? Assume that the tube is 'ideal' i.e. it has no dark current;
- What sources of interference from the lab environment should you worry about?

For more information on the Photomultiplier Tube see the handout or Diefenderfer and Holton chapter 7.

Note, however, that the setup described here is slightly different from that given in Diefenderfer figure 7.18

5.

- A metal ruler is marked on one side in metric units (to 30 cm) and the other side in Imperial units (to 12 inches). The metric side is graduated in increments of 0.5 mm and the Imperial side in 64^{th} of an inch. Express the *dynamic range* of both sides. Given $1\text{ m}=39.37\text{ inches}$, which side provides the highest resolution measurement?
- The micrometer is a tool for measuring the length of small mechanical parts. A regular micrometer (below, left) measures to a resolution of 0.01 mm and can accommodate an object up to 5 cm long within its jaws. A Vernier micrometer (right) has an additional scale on the barrel which allows a factor of 10 greater resolution. Express the dynamic range of both types in dB. This device clearly measures with a higher resolution compared to the ruler, but how long would our metric ruler have to be in order to have the same dynamic range as the Vernier micrometer?



(c) A metal rod about 10 cm long is passed around to several groups of students. Each group has five members and each member makes an independent measurement of the rod length. Each group uses a different measurement instrument to do the job. Their results are presented in the table below (all in cm).

1. Which group has the highest resolution measurement device? And the lowest?
2. Which group has the most precise measurement?
3. Which group has the most accurate measurement?
4. Which group has the worst repeatability?

Member	#1	#2	#3	#4	#5
Group A	10.1	10.4	9.6	9.9	10.8
Group B	10.135	10.227	10.201	10.011	10.155
Group C	12.14	12.17	12.15	12.14	12.18
Group D	10.05	10.82	8.01	11.5	10.77
Group E	10	11	10	10	10

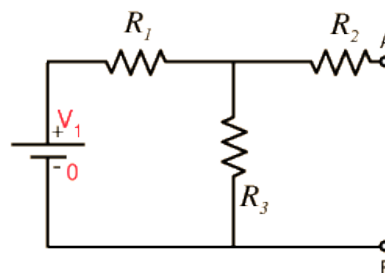
We now receive a report from the workshop where the rod was manufactured stating that the rod is in fact 101.600 ± 0.001 mm long. They have measured this with a system fully calibrated with paperwork traceable to a national standard, so we can believe this. Given this new information, answer the following questions:

1. Which group has the best precision? Has your answer changed from (2) above?
2. Which group has the largest error? Is this systematic or random? Explain your reasoning.
3. Which group has the most accurate measurement?
4. Which group has the least precise measurement?

6.

(a) The circuit in the figure consists of an ideal voltage source and three resistors. Draw a diagram of the Thevenin equivalent circuit.

(b) For $V_1 = 10$ V, $R_1 = 1$ k Ω , $R_2 = 2$ k Ω , $R_3 = 3$ k Ω find the Thevenin equivalent voltage V_{Th} and the Thevenin equivalent resistance R_{Th} .



(c) Draw the Norton equivalent circuit.

(d) Using the same values as in (b), find the Norton equivalent resistance R_N and equivalent current I_N .

7.

(a) Draw the Thevenin equivalent circuit for the LM35 temperature sensor at a temperature of 100°C.

(b) What voltage would be measured with a voltmeter of input resistance

- i) 1 Ω
- ii) 1 k Ω
- iii) 1 M Ω

(c) Draw the Norton equivalent circuit for the AD590 temperature sensor at a temperature of 300 Kelvin.

(d) What current would be measured with an ammeter of input resistance

- i) 1 Ω
- ii) 1 k Ω
- iii) 1 M Ω

(e) Which of the above values of resistance is appropriate for a voltmeter? And for an ammeter? Justify.

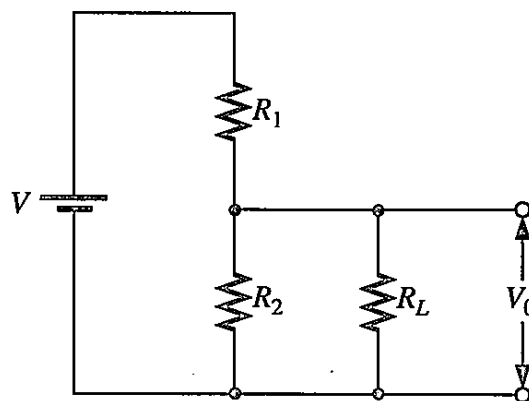
8.

(a) The circuit shown below is a 'voltage divider' (which may represent some voltage which we wish to measure) connected to a load resistance R_L (which may represent the measuring device). With R_L removed from the circuit, redraw this as a Thevenin equivalent and find the Thevenin voltage and resistance.

(b) Hence show that, when R_L is re-connected, the voltage divider is 'loaded' and the measured voltage V_o becomes

$$V_o = \frac{VR_2}{\left(\frac{R_1 R_2}{R_L}\right) + R_1 + R_2}$$

(c) For which condition does the measured voltage tend to the Thevenin voltage?



Instrumentation

Some numerical answers and hints:

2.

(a) The simplest geometry is to assume that the mercury reservoir bulb is a sphere with radius L .

(b) The volume change calculated in (a) is the volume swept-out as the mercury moves up the bore, which can be stated in terms of the length and radius of the bore

(c) 0.4 mm

3.

2.8 seconds (2 time constants).

In addition to the settling-time error, look for non-linearity error and uncorrected calibration error.

Combine error sources as root sum of squares to get 2.2°C

4.

(b) 9.4×10^{-13} C

(c) 2.4×10^{-10} W

5.

(a) Metric dynamic range = 600, Imperial dynamic range = 768

(b) 74 dB, 94 dB, 25 m

6.

(b) 7.5V and 2.75 k Ω (d) 2.73 mA and 2.75 k Ω .

7.

(b) 0.91 V, 1 V, 1 V (d) 300 μ A, 299.98 μ A, 272.73 μ A (all to 2 decimal places).